

ACEH OCEAN INTELLIGENCE BRIEF

Extending Ocean Understanding with AI (2024–2026)

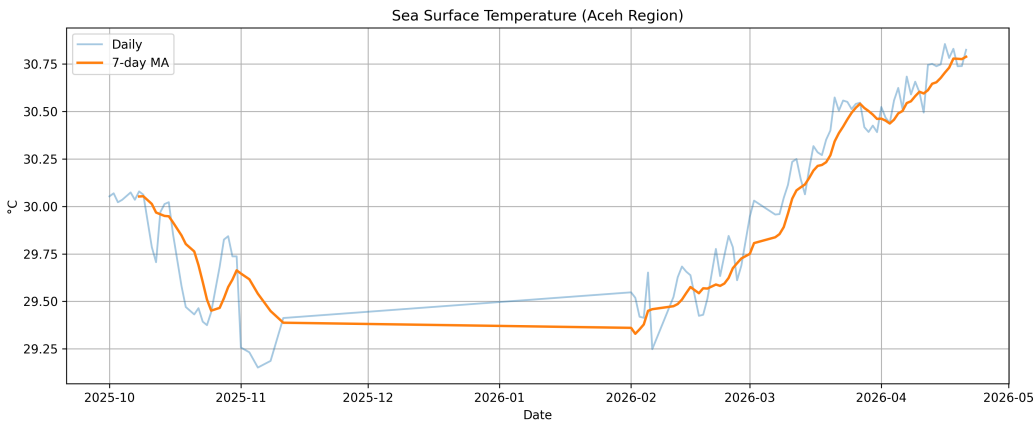


Figure 1. Sea Surface Temperature with 7-day moving average.

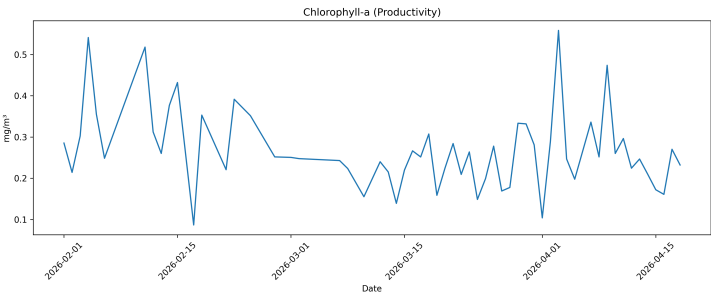


Figure 2. Chlorophyll-a variability indicating episodic productivity.

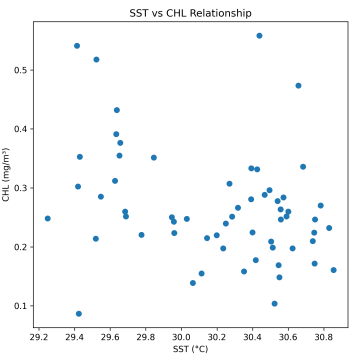


Figure 3. Non-linear SST–CHL relationship.

Overview

This brief presents an AI-assisted extension of Copernicus Marine data for the Aceh region, transforming observational data into operational ocean intelligence.

Key Observations

- SST shows a persistent warming trend from February to April 2026.
- Chlorophyll-a remains highly variable and episodic.
- The SST–CHL relationship is dispersed and non-linear.

Scientific Insight

Marine productivity in the Aceh region is not directly governed by surface temperature alone, but by dynamic processes such as vertical mixing, nutrient transport, and episodic forcing.

System Interpretation

The Aceh marine system exhibits non-linear behavior, indicating a complex interaction between physical and biological processes.

Positioning

This brief demonstrates how Copernicus Marine data can be translated into a real-world, AI-assisted ocean intelligence workflow for coastal understanding and scientific communication.